

Savitribai Phule Pune University, Pune
Sharadchandra Pawar College Of Engineering
Department of Computer Engineering

A Presentation
on

**“AI-Based Smart Transportation and Fleet Management Platform with Live
Vehicle Tracking, Driver Behavior Analysis, Fuel Optimization and Accident
Prediction ”**

Presented By

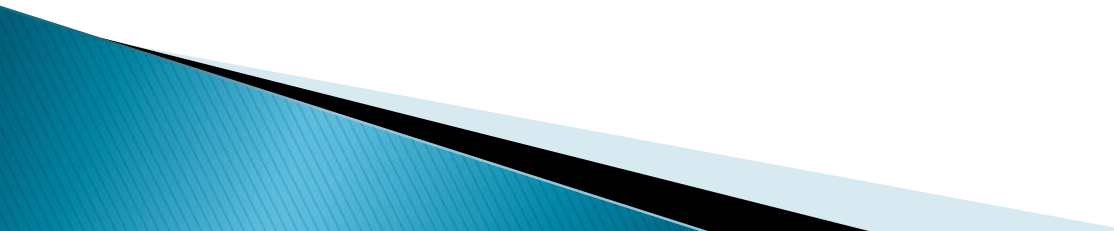
- 1. BRAMHANE ROHINI GANESH**
- 2. TAMBOLI ALISHA SULEMAN**
- 3. SHIMPLE SHAMAL ABHIMANYU**
- 4. RAUT ADITI SHIRISH**

B.E (Computer)

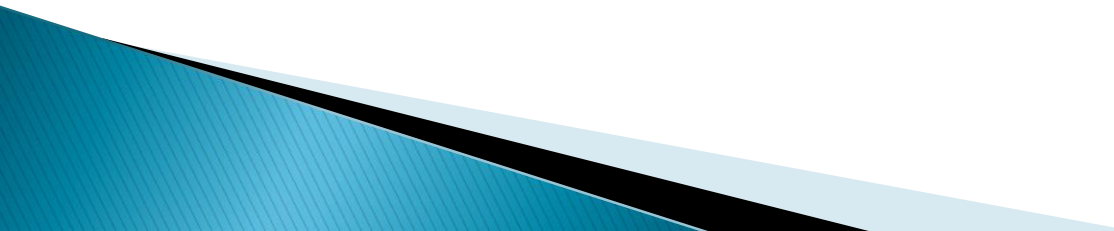
Under the Guidance of

Dr Khatal S.S

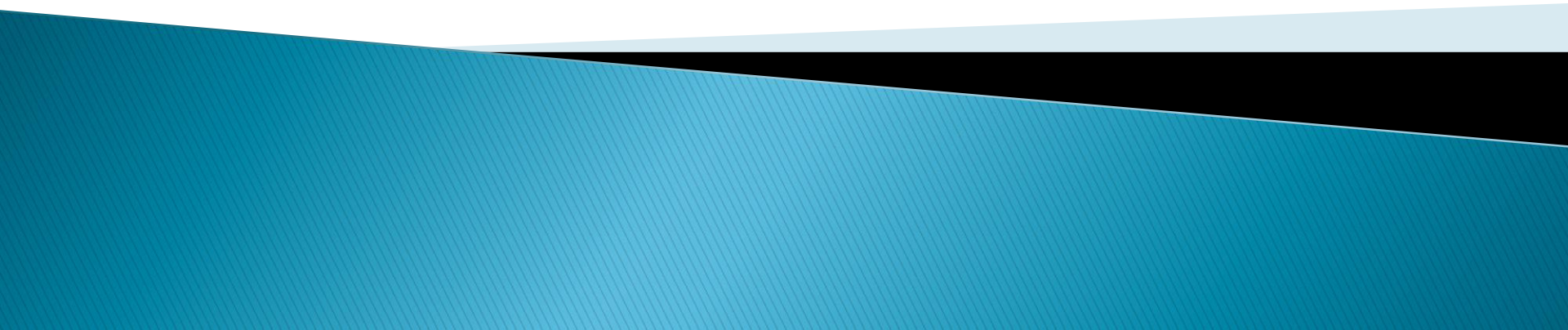
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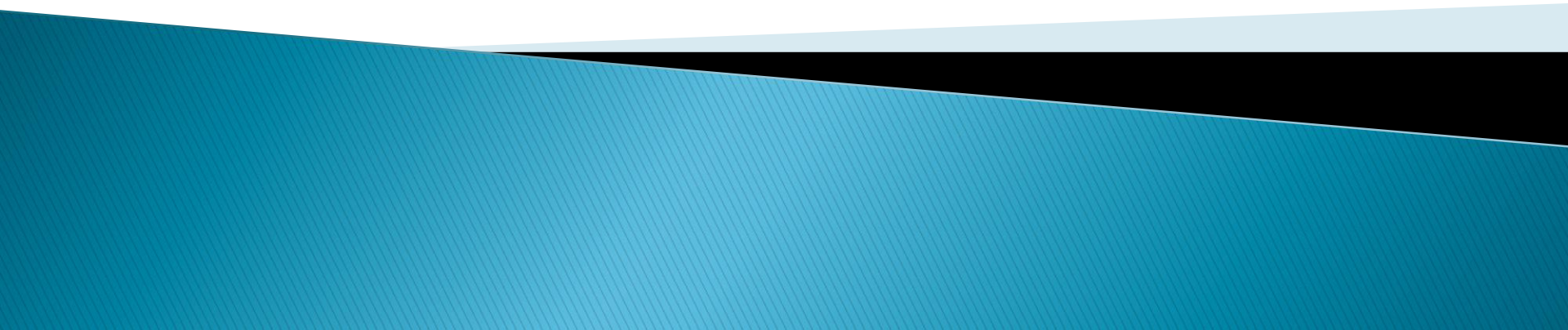
Introduction

- ▶ Modern transportation systems generate large amounts of vehicle and driver data. Data includes location, speed, acceleration, braking, journey details, fuel consumption and driver performance. Traditional fleet analysis mainly depends on spreadsheets, historical reports and manual examination. Manual analysis makes it difficult to identify risky driving patterns and inefficient fuel usage. AI and Machine Learning can identify hidden patterns and provide useful predictions. The proposed system converts historical transportation data into actionable fleet-management insights.
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Problem Statement

- Fleet data is often analyzed separately using spreadsheets or static reports. Large datasets are difficult and time-consuming to analyze manually. Risky driving patterns may remain undetected. Accident-prone journeys are difficult to identify from historical records. Fuel inefficiencies may not be clearly visible. Vehicle, driver, risk and fuel information are not integrated into one platform.
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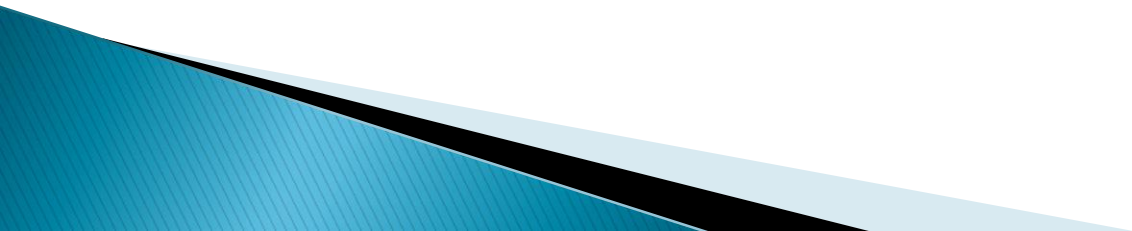
Motivation

- Need for efficient analysis of large fleet datasets.
 - Limitations of traditional spreadsheet-based analysis.
 - Need to identify normal and risky driver behavior.
 - Need for accident-risk prediction.
 - Need to identify inefficient fuel-consumption patterns.
 - AI models can discover relationships in sequential transportation data that are difficult to detect manually.
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Literature survey

Title	Year	Author	Short Summary
Driving Behavior Recognition using Deep Learning	2025	H. Li et al.	Uses AI and deep learning to identify risky driving behavior.
Driving Risk Assessment using Deep Learning	2025	W.-H. Lee & C.-Y. Chang	Uses LSTM, Bi-LSTM and GRU to predict driving risk.
Driver Behavior and Fuel Consumption Prediction	2019	P. Ping et al.	Studies how driver behavior affects vehicle fuel consumption.
Driving Risk Prediction using XGBoost	2019	X. Shi et al.	Uses XGBoost to analyze driving behavior and predict safety risks.
Driving Safety Risk Prediction using Data	2018	N. Arbabzadeh & M. Jafari	Uses driver and road information to predict driving safety risks.

Objectives

- To develop a CSV-based vehicle tracking and fleet monitoring system.
 - To analyze driver behavior using speed, acceleration, braking, and journey data.
 - To predict driver and accident-risk levels using AI/ML models.
 - To analyze fuel consumption and identify inefficient fuel usage.
 - To provide all vehicle, driver, risk, fuel, and fleet performance information through a centralized dashboard.
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Proposed Methodology

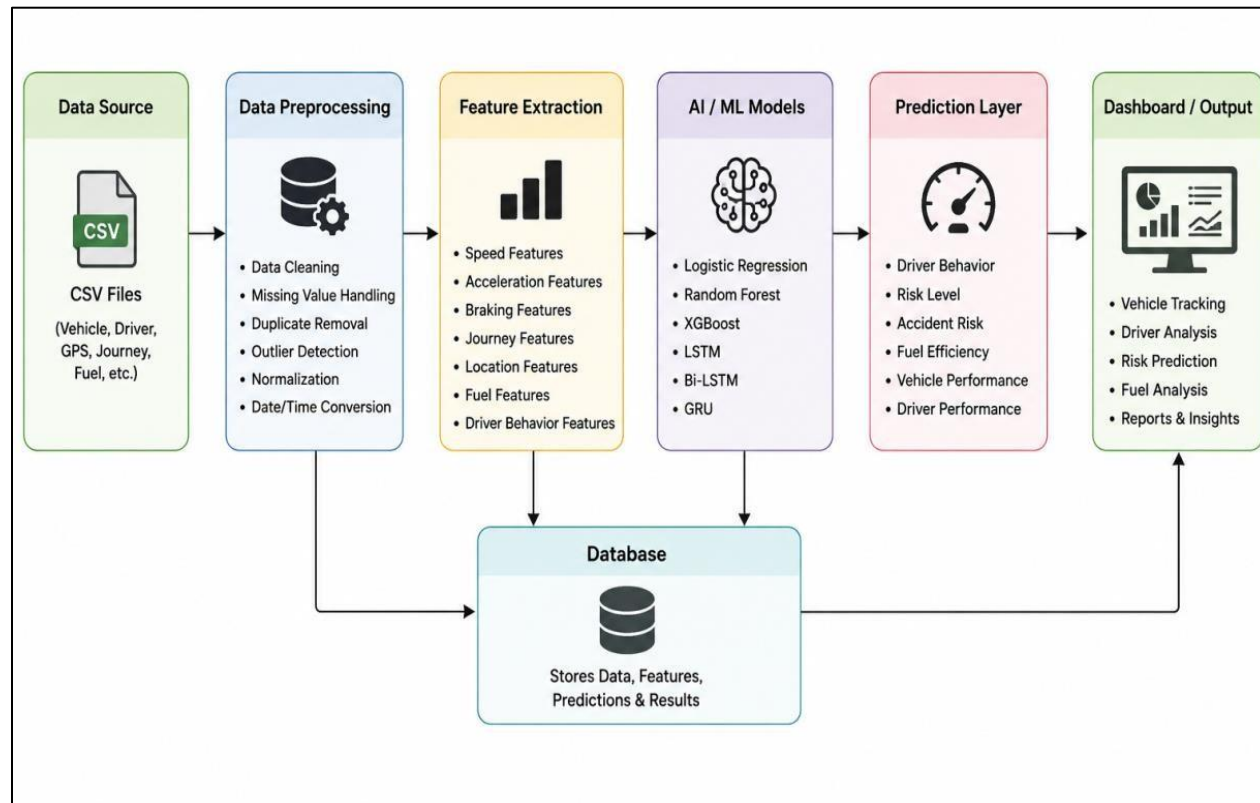


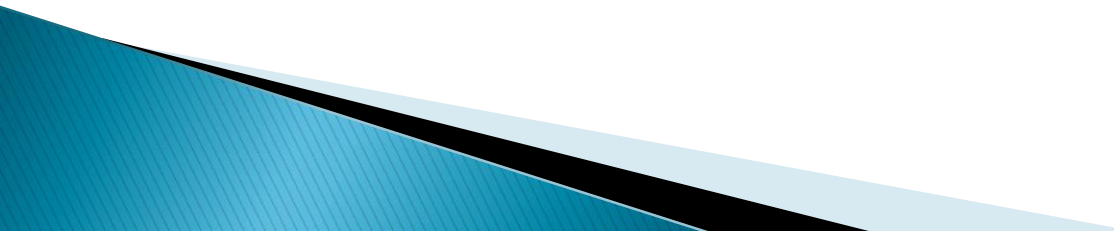
Fig. Block Diagram

Tools and Technology

Software Requirements

- ▶ **OS:** Windows 10
- ▶ **Language:** React , Python 3.12 , Fast AI , Machine Learning
- ▶ **Libraries:** Scikit-learn, TensorFlow / PyTorch, Pandas, NumPy, Matplotlib
- ▶ **Database:** SQLite
- ▶ **Tools:** VS Code

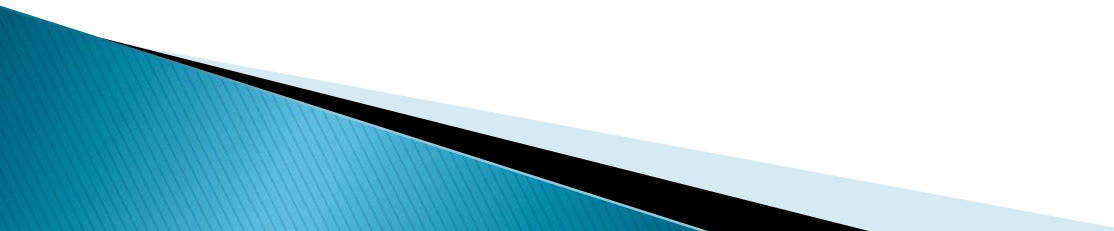
Hardware Requirements

- ▶ **Minimum:** Intel i5, 8 GB RAM, 256 GB SSD
 - ▶ **Recommended:** Intel i5 / Ryzen 5, 16 GB RAM, 512 GB SSD, NVIDIA GPU (RTX 1450 or higher)
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Expected Result

- The proposed system is expected to provide vehicle tracking and journey visualization using the available GPS and vehicle data. It will analyze driver behavior based on speed, acceleration, braking, and journey information to identify normal and risky driving patterns. The system will also predict driver and accident-risk levels using AI and machine-learning models. In addition, it will analyze fuel-consumption patterns to identify inefficient fuel usage and support fuel optimization. Finally, all the results related to vehicles, drivers, safety, fuel consumption, and fleet performance will be presented through a centralized dashboard for better fleet-management decisions.

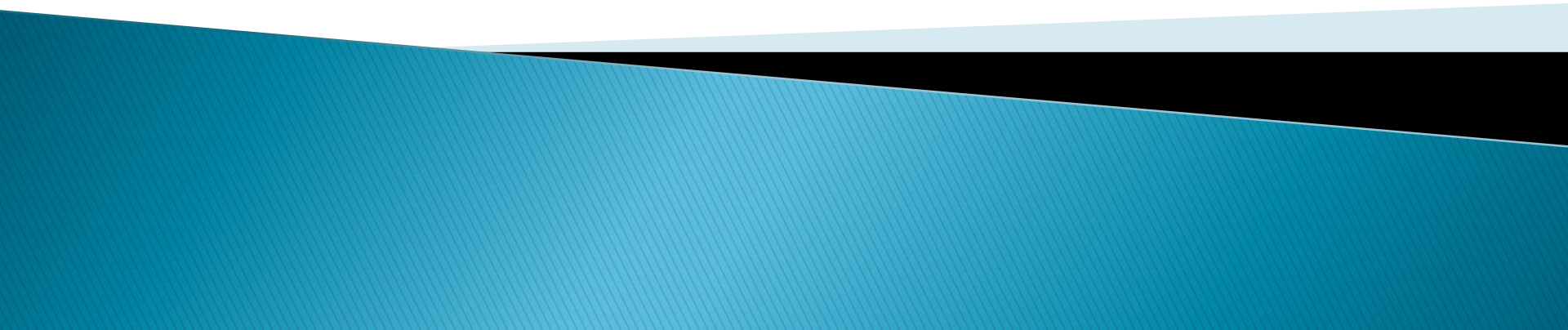
Scope of the work

- ▶ The scope of this project is to develop an AI-based smart transportation and fleet management platform using historical transportation data stored in CSV files. The system will process vehicle, driver, GPS, journey, and fuel-related data to perform vehicle tracking, driver behavior analysis, accident-risk prediction, fuel-efficiency analysis, and fleet performance evaluation. Machine Learning and Deep Learning models such as Logistic Regression, Random Forest, XGBoost, LSTM, Bi-LSTM, and GRU will be evaluated. The analyzed results will be integrated into a centralized dashboard to help fleet administrators monitor vehicles, identify risky driving patterns, improve fuel efficiency, and make data-driven fleet-management decisions.
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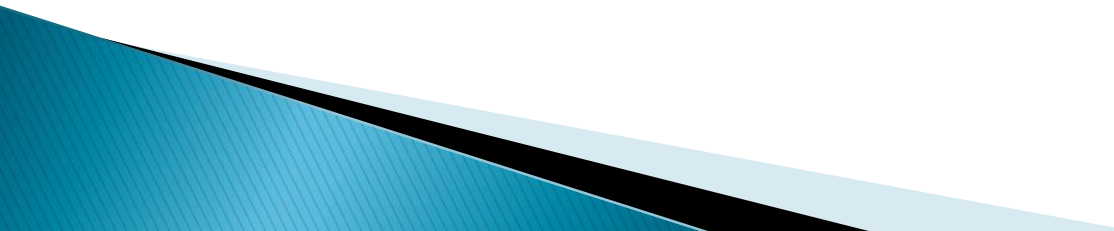
Work Plan

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Conclusion

- The proposed AI-Based Smart Transportation and Fleet Management Platform provides an efficient way to analyze historical transportation data and improve fleet management. The system combines vehicle tracking, driver behavior analysis, accident-risk prediction, fuel optimization, and fleet performance analysis using Machine Learning and Deep Learning techniques. The centralized dashboard will present these results in an easy-to-understand manner, helping fleet administrators make safer, more efficient, and data-driven decisions.
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Future Scope

- ▶ In the future, the proposed system can be extended from CSV-based historical data to real-time IoT and sensor data. Live GPS tracking and vehicle sensors can be integrated for continuous monitoring of speed, acceleration, braking, and fuel consumption. The system can also be enhanced with more advanced AI models for real-time accident prediction, driver-risk detection, and fuel optimization. Further improvements can include mobile applications, real-time alerts for high-risk driving, and advanced fleet-performance analytics.
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Thank You!!!